

Using the Climbing Wall Force Dashboard

Getting-started guide — connecting, recording, and reading the data.

The climbing wall is fitted with four force sensors — one under each hand hold and each foot hold. This web dashboard connects to the sensors, shows the forces in real time, and lets you record sessions, run a jump-height test, and compare past recordings. This guide shows you how to start the system and how to use each feature. It assumes only that you are comfortable opening a terminal and a web browser; no specialist knowledge is required.

1. Starting the system

The system has three parts that work together: the four **sensor boards** (one per hold, plugged into the laptop over USB), a **backend** (a small Python server that reads the boards, streams their data, stores recordings and does the jump calculation) and a **frontend** (the dashboard you see in the browser). Set them up in the order below — the two software parts each in their own terminal window. To explore in **demo mode** you can leave the sensor boards unplugged — the backend and frontend alone are enough.

1 — Plug in the sensor boards (live data only)

The four sensor boards connect to the laptop through a **USB hub**: each board's USB cable goes into the hub, and the hub plugs into the laptop with a single cable. That is all — there is nothing to log into and nothing to start on the sensor side. The backend (next step) reads the boards directly over USB.

First time on a new computer: the Phidget sensor driver must be set up once. On **macOS**, allow it under System Settings → Privacy & Security (click Allow next to the Phidgets driver). On **Windows**, install the Phidget22 drivers from phidgets.com/docs/OS_-_Windows. Without the driver the backend cannot open the boards (every channel fails with “device is in use”).

Boards may be plugged in or re-plugged at any time, before or after the backend is started — they are picked up automatically within a second or two. Only if you restart the *backend* itself do you need to re-click Connect Device on the website afterwards.

2 — Start the backend (Python)

In a terminal:

```
» cd backend
» python3 -m venv venv (first time only)
» source venv/bin/activate (Windows: venv\Scripts\activate)
» pip install -r requirements.txt (first time only)
» python app.py
```

You should see Running on `http://127.0.0.1:5001`. Leave this window open. There is also a one-step launcher in the **backend** folder that even does the first-time setup for you: **run_backend.bat** on Windows (double-click it, or run it from a terminal), **./run_backend.sh** on macOS/Linux. Use the one that matches your computer — a **.sh** script will not run on Windows (Windows asks “Open with?” instead) and a **.bat** will not run on macOS.

Run only one backend at a time. Each sensor board can be opened by exactly one program, so a second backend instance — or the Phidget Control Panel app left open — will fight over the boards and neither will get data.

3 — Start the frontend (Node)

In a second terminal:

```
» cd climbing_wall_website
» npm install
» npm run dev
```

(first time only)

This prints a local address such as `http://localhost:5173`.

4 — Open the dashboard

Open the address printed above in a web browser. Any modern desktop browser works (Chrome, Edge, Firefox, Safari) — the dashboard receives the sensor data from the backend, so it no longer needs any special browser feature. Without the sensor boards it still runs in **demo mode** with simulated data — handy for trying things out.

2. The dashboard at a glance

When the page loads you see the toolbar across the top, a status bar, and four tabs (Figure 1): the three described in this guide, plus a **Jump Test (BETA)** tab that is still experimental and not covered here.

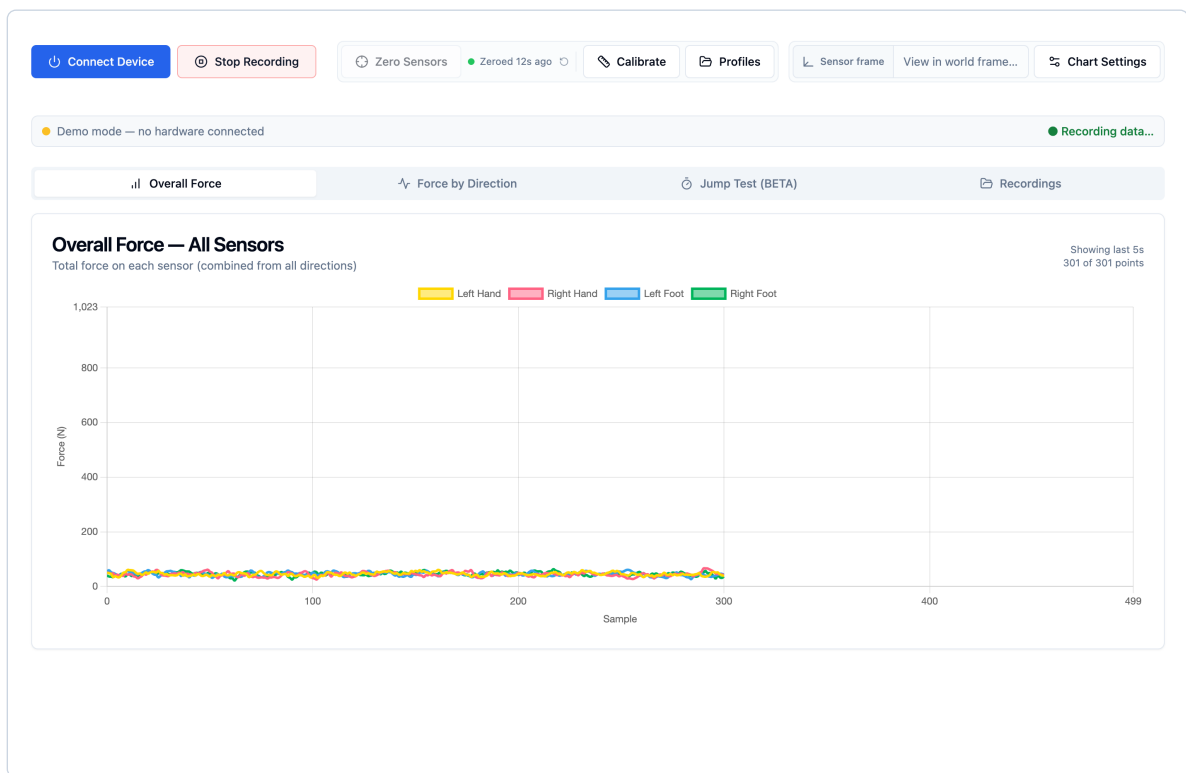


Figure 1. The dashboard: toolbar (top), status bar, the four tabs, and the *Overall Force* chart. Here the system is in demo mode and recording.

The toolbar. The controls are grouped into three trays that read left to right:

- **Session** — Connect Device opens or closes the live sensor stream from the backend (it turns into Disconnect once connected), and Start Recording / Stop Recording begins or pauses collecting data. While paused you also get Start New Recording, Save and Export Data.
- **Sensor calibration** — Zero Sensors sets the live zero (the “tare”), Calibrate opens the calibration wizard, and Profiles saves or loads named calibrations. All three are covered in the separate *Calibration Guide*.
- **View** — Sensor frame / View in world frame switches the coordinate system used for the charts (Section 5), and Chart Settings adjusts how the charts look (Section 6).

The status bar. A coloured dot reports the device: **green** = connected, **amber** = demo mode, **red** = no device. On the right it shows whether data is currently being recorded.

3. Recording a session

1. Click Connect Device to open the live sensor stream (skip this to stay in demo mode). If a warning appears saying only some of the 12 sensor channels are attached, check the USB cables of the sensors it names — missing channels read 0.
2. Click Start Recording. Live data begins to flow into the charts and the status bar shows “Recording data...”.
3. Climb, hang, or push on the holds. The charts update in real time.
4. Click Stop Recording to pause. Your data is kept.
5. Click Save to store the session on the backend (give it a name so you can find it later), or Export Data to download it as a .csv file you can open in Excel.
6. Click Start New Recording to clear the buffer and begin a fresh session.

The dashboard auto-saves the live buffer to the browser every few seconds, so a page reload loses at most a few seconds of data. “Save” (to the backend) and “Export” (to a file) are how you keep a session permanently.

4. The chart tabs

4.1. Overall Force

The first tab (Figure 1) shows the **total** force on each of the four sensors as one line each. “Total” means the combined strength of the pull regardless of direction (the length of the force vector, $\sqrt{X^2 + Y^2 + Z^2}$). Use it for a quick “how hard is each limb pulling?” overview. Because it ignores direction, this chart looks the same in either coordinate frame.

4.2. Force by Direction

The second tab (Figure 2) breaks each sensor’s force into its three components — *X*, *Y* and *Z* — on a separate small chart per sensor. This is where you see, for example, how much a hand is pulling *down* versus *out of the wall*. The toggle chips above the charts let you show or hide individual sensors, and each chart can be expanded.

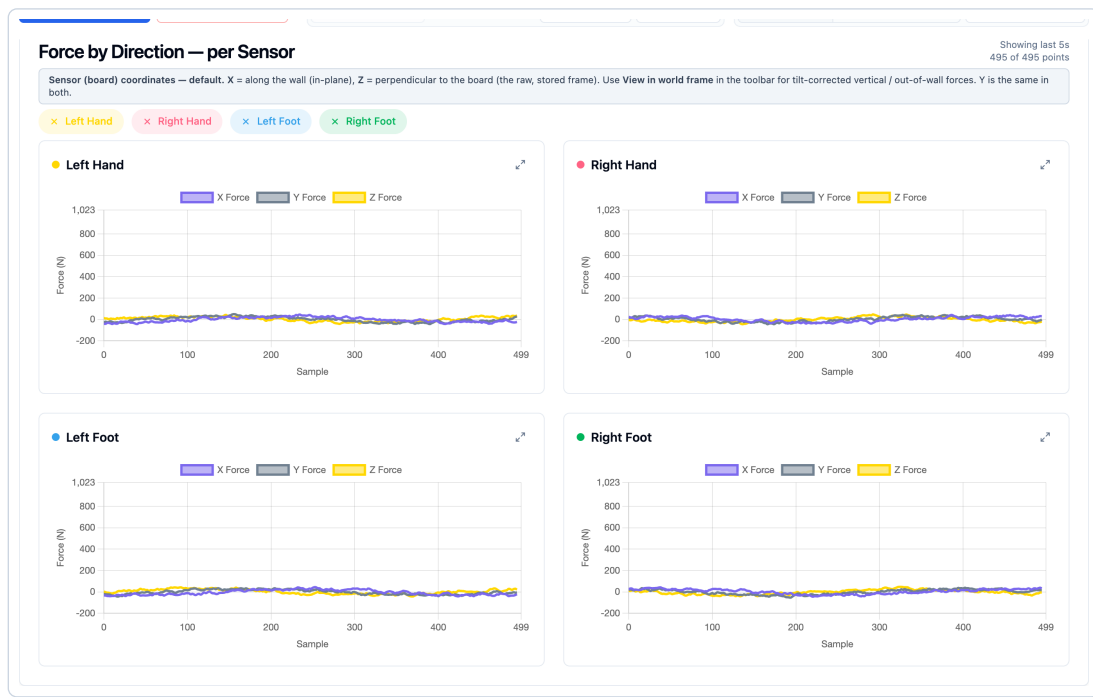


Figure 2. *Force by Direction*: each sensor's *X* (purple), *Y* (slate) and *Z* (gold) components over time.

4.3. Recordings

The *Recordings* tab (Figure 3) lists your **five most recent** saved sessions — older ones are not shown, but they are not lost either: every save stays on disk in the backend's **saved_recordings** folder. Click a recording to plot it: you get its *Overall Force* chart and the per-sensor *Force by Direction* charts. From here you can:

- toggle each recording between **Sensor** and **World** view (the world angle is taken automatically from the tilt stored with the recording);
- Export CSV to download the session;
- Compare two recordings side by side (available once you have at least two saved).

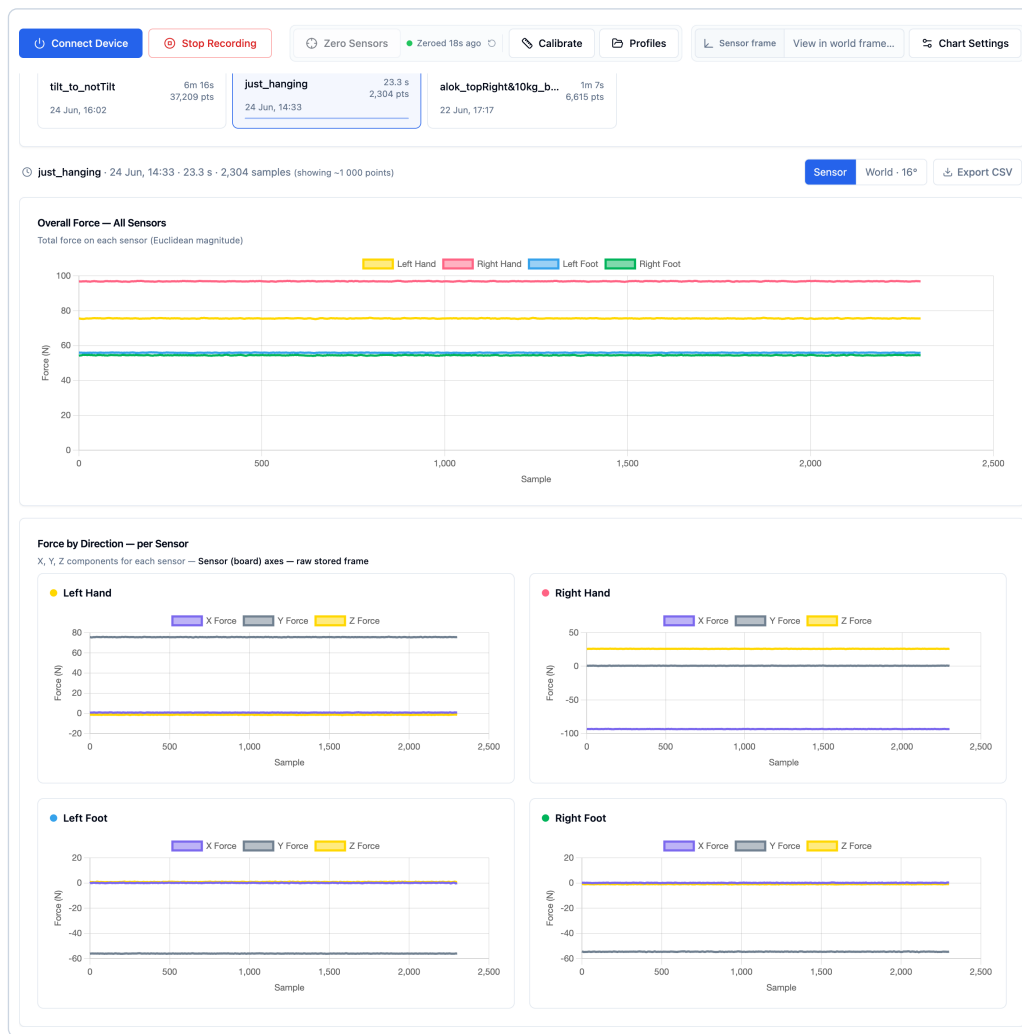


Figure 3. The *Recordings* tab showing a saved climbing session. The clear spike is a hard pull on one hold.

5. Sensor frame vs. world frame

The sensors are bolted to the (tilted) wall, so by default they report forces in the **sensor frame**: X runs *along* the wall surface and Z runs *perpendicular* to it. Often you would rather think in real-world terms — straight up versus straight out. The **world frame** rotates the X and Z components by the wall's tilt angle θ so that:

$$X \rightarrow \text{true vertical}, \quad Z \rightarrow \text{true horizontal (out of the wall)}, \quad Y \rightarrow \text{unchanged}.$$

Geometrically the two frames share a single origin and differ only by the tilt: the world frame is the sensor frame rotated by θ (Figure 4).

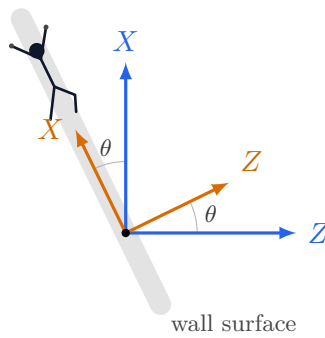


Figure 4. Sensor frame vs. world frame, seen side-on, with a climber on the wall to fix the geometry. The **sensor frame** is bolted to the board, so its *X* runs *along the wall surface* the climber is on and its *Z* points straight out of it. The **world frame** instead has *X* truly vertical and *Z* truly horizontal — it is just the sensor frame rotated by the wall tilt θ . The *Y* (sideways) axis points out of the page and is the same in both frames.

To switch the live view, click *View in world frame...*, type the *current* wall tilt, and confirm (Figure 5). The view locks at that angle and can be unlocked at any time. The tilt you type here is a **display-only** angle, separate from the calibration angle θ set in the calibration wizard: it is merely pre-filled from it, and changing it never touches the calibration.

View in world frame

Enter the **current** wall tilt to rotate X/Z into true vertical / out-of-wall. This only changes the display — stored data stays in the sensor frame.

Current wall tilt θ (–90° to 90°)

16

° from vertical

Cancel

View in world frame

Figure 5. Opting into the world-frame view: enter and confirm the current wall tilt θ .

The world frame only changes *what the charts show*. The stored data always stays in the raw sensor frame, so switching back and forth never alters or damages a recording. The *Overall Force* chart is unaffected because total force does not depend on direction.

6. Chart settings

The Chart Settings button opens a small panel (Figure 6) with three controls:

- **Time window shown** — how much recent history to display (e.g. the last 5 seconds, or everything).
- **Force range (chart top)** — the maximum value on the vertical axis, so you can zoom in or out on the force scale.
- **Auto-adjust scale to data** — let the chart pick the vertical range automatically.

Chart Settings

Time window shown

How much history to display on the chart.

Last 5 seconds

Force range (chart top)

1023 N

☐ Auto-adjust scale to data

Data points recorded

1,053

Figure 6. *Chart Settings:* time window, force range and auto-scaling.

7. Calibration and zeroing

Before the forces shown are accurate in newtons, each board must be **calibrated** against known weights — a one-time procedure done from the **Calibrate** button. Separately, you **zero** the sensors at the start of each session with the **Zero Sensors** button, which re-sets the baseline so an unloaded hold reads zero. Calibration is rare and saved; zeroing is quick, live, and done often. Both are important enough to have their own document — please see the separate **Calibration Guide** for the full walkthrough (including saved Profiles) and the reasoning behind it.

8. Troubleshooting

Problem	Fix
Connect Device fails or falls straight into demo mode	Make sure the backend is running (<code>python app.py</code> in the backend terminal) — the live stream comes through it.
A warning says only some of the 12 sensor channels are attached	One or more boards are unplugged. Check the USB cables into the hub for the sensors named in the warning — re-plugged boards are picked up automatically within a second or two.
Boards are plugged in but no channels attach	Allow the Phidgets driver once (macOS: System Settings → Privacy & Security) and restart the backend. Also close the Phidget Control Panel and any second backend instance — each board can only be opened by one program at a time.
“No backend” / recordings will not save	Make sure <code>python app.py</code> is still running in the backend terminal.
node or python3 “command not found”	Install Node.js (LTS) from nodejs.org or Python from python.org .
The charts are empty	Press Start Recording . With no device connected the system shows demo data so you can still explore.
Forces look shifted, or too large or too small	First Zero Sensors with nothing on the holds. If they are still off, the sensors probably need calibrating — see the Calibration Guide.

Screenshots were taken from the dashboard running in demo mode with simulated data.